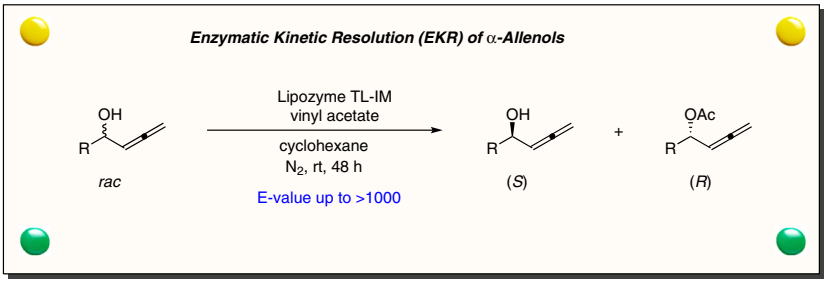
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Enzymatic Kinetic Resolution (EKR) of α-Allenols Enabled by a Commercially Available Immobilized Lipase

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Published as part of the Special Issue
Thieme Chemistry Journals Awardees 2023



Received: 14.09.2023
Accepted after revision: 30.10.2023
Published online: 27.11.2023 (Version of Record)
DOI: 10.1055/s-0042-1751525; Art ID: ST-2023-09-0402-L

Abstract Enzymatic kinetic resolution (EKR) of racemates has been considered as an efficient and scalable technology to deliver enantiopure substances in an environmentally benign way. Herein, we develop a practical EKR of α-allenols enabled by a commercially available immobilized lipase, Lipozyme TL-IM. The stereoselectivity factor (*S*-factor) of this kinetic resolution (KR) was investigated to reach up to >1000. The robust nature of the EKR strategy is reflected by a broad scope of α-allenol substrates with the excellent control of enantioselectivity. This method offers an alternative approach to access chiral α-allenols, which can be easily transformed into many valuable oxycycles.

Keywords enzymatic kinetic resolution (EKR), α-allenols, lipase, kinetic resolution, enzymes

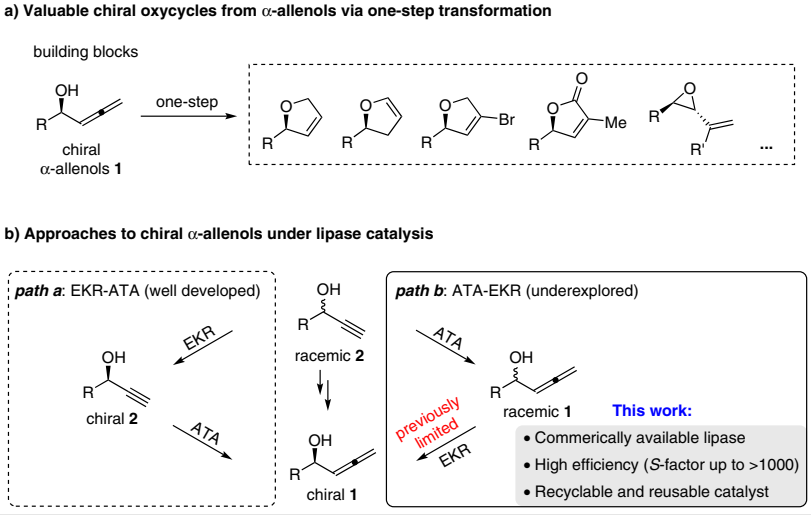
Enzymatic kinetic resolution (EKR) is one of the most practical and industrially widespread tools in the separation of enantiomers.[1,2] In particular, lipase-catalyzed kinetic resolution (KR) has been demonstrated as an efficient and scalable approach to produce chiral secondary alcohols, and its use of less toxic solvents and enzymes presents as an environmentally benign and atom economical approach to produce optically active compounds.[3] On the other hand, chiral α-allenols have been widely used as building blocks in synthetic chemistry, and it can also be easily converted into diverse and valuable chiral oxycycles (Scheme 1a).[4,5] Previous methods to chiral α-allenols were mainly focused on the allenation of terminal alkynes (ATA)[6,7] starting from chiral propargylic alcohols, which could be conveniently obtained from racemic propargylic alcohols via EKR (path a, Scheme 1b).[8] Direct resolution of racemic α-allenols presents as an alternative and practical method with the formation of enantiopure compounds at a late stage, which can improve the tolerance of functional groups, and avoid the possible loss of enantiopurity during the late-stage allenation step, generally requiring relatively harsh conditions.[7] However, very limited approaches were realized for the direct EKR of α-allenols, given the fact that the bulky allene unit does not fit well as the medium-sized group into the active site of typical lipases, thus leading to low enantioselectivity.[9,10] In 2015, the Bäckvall group described an enantioselective access to 2,3-dihydrofurans from α-allenols by cascade catalysis, where homemade IL1-PS realized the EKR of racemic α-allenols in the first step.[11] It is noteworthy to mention that the use of commercially available catalyst endows the developed methods with convenience and higher possibilities for potential applications in other laboratories and industry as well. Furthermore, immobilization of the lipase catalyst would highly improve its practical application value in industry, due to the recyclability and reusability. In 2016, Hong, Huang, and co-workers elegantly reported commercially available lipase AK-catalyzed direct and efficient EKR of α-allenols, while, relatively low enantioselectivity was observed for electron-rich substrates.[12] Herein, we report our recent observations on the EKR of α-allenols enabled by a commercially available immobilized lipase, Lipozyme TL-IM, exhibiting high efficiency and selectivity in the resolution of α-allenols.

Preparation of Racemic α-Allenols *rac*-1a–m via ATA Reactions

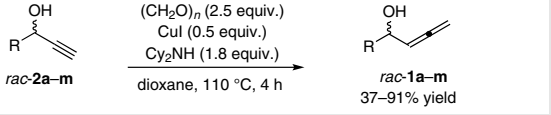
Our work began with the synthesis of racemic α-allenols *rac*-**1** via one-step ATA reactions.[7b] Under the standard conditions, *rac*-**1a–m** could be conveniently obtained from propargylic alcohols *rac*-**2a–m** in 37–91% yields (Scheme 2).

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Scheme 1 State-of-the-art and this work



Scheme 2 ATA approach to access racemic α-allenols from propargylic alcohols

Optimization of the EKR Reaction Conditions

Next, a collection of biocatalysts was studied for the EKR of α-allenol **1a** with vinyl acetate as acyl donor in toluene at 35 °C for 14 h, and *S*-factor was calculated to evaluate lipase performance in the catalytic esterification (Table 1). The reaction with Lipozyme TL-IM gave the highest *S*-factor (637), which is almost three times as high as *S*-factor (242) from the reaction catalyzed by lipase AK. Lipozyme TL-IM, originating from *Thermomyces lanuginosus*, is a commercially available and low-cost lipase, which is immobilized on a noncompressible silica gel carrier. To our delight, reaction with lower lipase loading gave even higher enantioselectivity, and cyclohexane was suggested to be the best solvent to deliver >99% *ee* for both ester product (*R*)-**3a** and recovered (*S*)-**1a**, exhibiting >1000 of *S*-factor at this stage. By prolonging the reaction time, lipase loading could be decreased to only 0.2 w/w, producing the same level of reaction enantioselectivity. Therefore, Lipozyme TL-IM (0.2 w/w), vinyl acetate (2.0 equiv.), in cyclohexane at rt for 48 h were defined as the optimized reaction conditions for further study.

Substrate Scope of α-Allenols

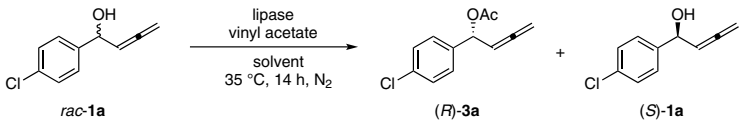
The scope of the EKR reaction was then investigated by using the optimal reaction conditions in Table 1 (Scheme 3).

We first studied the reactivity of α-allenols with substituent at the *para* position on the benzene ring, such as chloro, bromo, or trifluoromethyl group, and they are all successfully converted into the corresponding optically pure esters kinetically, providing >1000 of *S*-factor in all cases. Reaction enantioselectivity was decreased for α-allenol **1d**, bearing one chloro group at the *meta* position. It is noteworthy to mention that the esterification step became much slower with a substituent at the *ortho* position due to the steric effect, and in this case Na_2CO_3 was added to dramatically accelerate the kinetic esterification. The benzyl group can also be tolerated, as shown by the reaction outcomes of **1f**. Since low enantioselectivity was observed for electron-rich substrates catalyzed by lipase AK in the previous report,[12] we subsequently focused on α-allenols bearing electron-donating groups (EDG). The 4-methoxy group on the benzene ring led to a decreased *S*-factor in the EKR process, but still almost three times higher than that catalyzed by lipase AK. EKR performances by lipozyme TL-IM are significantly improved for α-allenol substrates bearing one or two 3-methoxyl group(s). Bulky groups are greatly challenging to fit the large pocket of lipase, thus generally resulting in low enantioselectivity for such kinds of substrates. However, to our delight, EKR of α-allenol **1j**, bearing 2-naphthyl group, gave equally excellent selectivity. An electron-donating bulky group leads to a decreased *S*-factor of 127 in the resolution of **1k**. Finally, by increasing the catalyst loading to 2.0 w/w, reaction of α-allenol **1l** with a bulky group of 1-naphthyl can finish in 48 h, affording chiral ester (*R*)-**3l** in 48% yield with 98% *ee*, and the recovered (*S*)-**1l** in 47% yield with 96% *ee*.

We next studied the reactivity and selectivity of α-allenol *rac*-**1m**, bearing a completely aliphatic side chain. As shown in Scheme 4, the kinetic resolution of *rac*-**1m** (1.0 mmol) under Lipozyme TL-IM produced the corresponding

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Table 1 EKR of Racemic α-Allenols: Conditions Optimization[a]



Entry	Lipase	Lipase loading (w/w)	Solvent	*ee* of 3a (%)[b]	*ee* of 1a (%)[b]	Conversion (%)[c]	S-factor[c]
1	CALB	1	toluene	97	55	37	114
2	lipase AK	1	toluene	>99	20	15	242
3	lipase RM	1	toluene	97	28	22	86
4	lipase NS	1	toluene	93	16	14	32
5	lipase AYS	1	toluene	87	11	12	16
6	lipase CL	1	toluene	–	–	N.R.	–
7	lipase PPL	1	toluene	–	–	N.R.	–
8	lipase PS-IM	1	toluene	80	>99	55	46
9	Lipozyme TL-IM	1	toluene	>99	91	48	637
10	Lipozyme TL-IM	0.5	toluene	>99	96	49	790
11	Lipozyme TL-IM	0.5	THF	94	5	5	209
12	Lipozyme TL-IM	0.5	DCM	>99	17	15	235
13	Lipozyme TL-IM	0.5	MeCN	>99	17	15	235
14	Lipozyme TL-IM	0.5	acetone	>99	5	5	209
15	Lipozyme TL-IM	0.5	cyclohexane	>99	>99	50	>1000
16	Lipozyme TL-IM	0.2	cyclohexane	>99	>99	50	>1000
17[d]	Lipozyme TL-IM	0.2	cyclohexane	>99	>99	50	>1000
18[e]	Lipozyme TL-IM	0.2	cyclohexane	>99	>99	50	>1000

[a] The reaction was carried out with α-allenol **1a** (0.2 mmol), lipase, and vinyl acetate (5.0 equiv.) in indicated solvent (1.0 mL) at 35 °C for 14 h; enantiomeric excess (*ee*) was determined by chiral HPLC.
[b] Enantiomeric excess (*ee*) was determined by chiral high-performance liquid chromatography (HPLC).
[c] The *S*-factor was calculated by *ee* of **3a** (ee_p) and *ee* of recovered **1a** (ee_s) according to the equation [$S = \ln[(1-ee_s)/(1+(ee_s/ee_p))]/\ln[(1+ee_s)/(1+(ee_s/ee_p))]$]. Conversion (*c*) is calculated according to the equation [$c = ee_s/(ee_s+ee_p)$].
[d] The reaction was carried out with α-allenol **1a** (0.2 mmol) and vinyl acetate (2.0 equiv.) at rt for 24 h.
[e] The reaction was carried out with α-allenol **1a** (1.0 mmol) and vinyl acetate (2.0 equiv.) at rt for 48 h. THF = tetrahydrofuran; DCM = dichloroethane; N.R. = no reaction.

ester (*S*)-**3m** in 35% yield with 88% *ee*, and (*R*)-**1m** was recovered in 61% yield with 48% *ee*. Therefore, the *S*-factor was observed to be 25 in this reaction.

With obtained optically pure α-allenols, we then tested synthetic applications of (*S*)-**1a** (Scheme 5). Silver-mediated cyclization of (*S*)-**1a** (>99% *ee*) delivered 2,5-dihydrofuran (*S*)-**4a** in 85% yield with 98% *ee*,[10a] while cycloisomerization of (*S*)-**1a** catalyzed by Shvo's catalyst afforded 2,3-dihydrofuran (*S*)-**4b** in 66% yield with >99% *ee*.[11] These chiral oxycycles are valuable building blocks in synthetic chemistry.

In conclusion, we have developed an efficient EKR method of α-allenols, realized by a commercially available immobilized lipase, Lipozyme TL-IM. The *S*-factor of this kinetic resolution method was investigated to reach up to >1000. The substrate scope has shown that not only electron-withdrawing groups, but also electron-donating groups could be nicely tolerated on the benzene ring. α-Allenols bearing a bulky group were also compatible to deliver chiral esters kinetically. This method offers an alternative approach to access chiral α-allenols and their esters with high enantioselectivity, which will be of high value for the synthesis of chiral oxycycles.

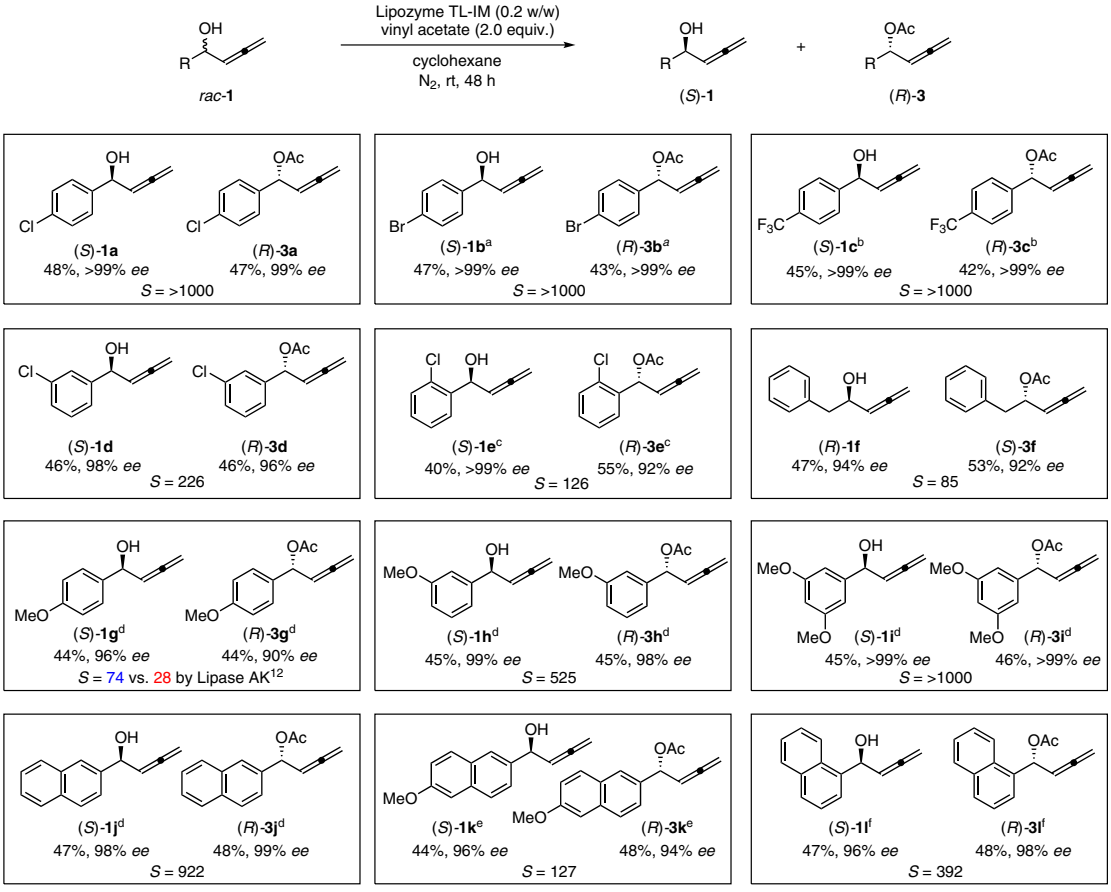
Conflict of Interest

The authors declare no conflict of interest.

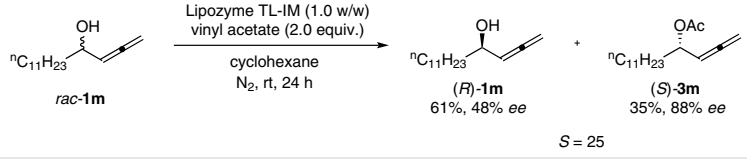
Funding Information

We acknowledge the National Natural Science Foundation of China (Grant No. 22271054), the Thousand Young Talents Program of China, and Fudan University (start-up grant) for financial support.

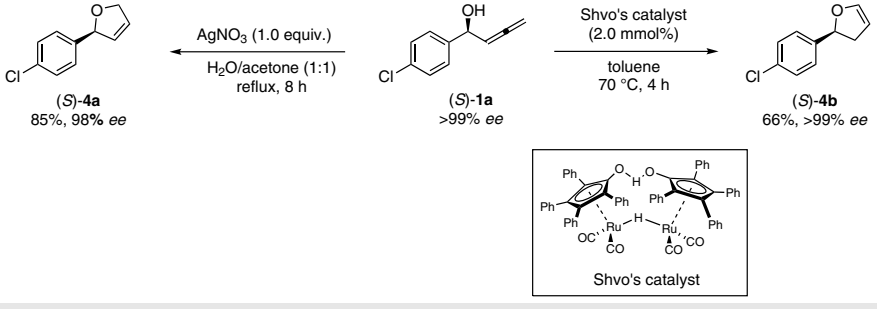
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Scheme 3 Scope of α-allenols for the EKR catalyzed by lipozyme TL-IM. The reaction was carried out with α-allenol **1** (1.0 mmol), vinyl acetate (2.0 equiv.), and Lipozyme TL-IM (0.2 w/w) in dry cyclohexane (5.0 mL) at rt for 48 h; enantiomeric excess (*ee*) was determined by chiral HPLC. The *S*-factor was calculated by *ee* of **3a** (ee_p) and *ee* of recovered **1a** (ee_s) according to the equation $\{S = \ln[(1-ee_s)/(1+(ee_s/ee_p))]\}/\ln[(1+ee_s)/(1+(ee_s/ee_p))]\}$. [a] The reaction time was 60 h. [b] The reaction time was 24 h. [c] The reaction was carried out with vinyl acetate (10 equiv.), Lipozyme TL-IM (2.0 w/w), and additional Na_2CO_3 (1.5 equiv.), while reaction conversion was observed to be only 32% in the absence of Na_2CO_3. [d] The reaction was carried out with Lipozyme TL-IM (0.5 w/w). [e] The reaction was carried out with Lipozyme TL-IM (1.0 w/w) in dry toluene (5.0 mL) at 40 °C for 48 h. [f] The reaction was carried out with vinyl acetate (10 equiv.) and Lipozyme TL-IM (2.0 w/w).



Scheme 4 EKR of *rac*-**1m** catalyzed by Lipozyme TL-IM



Scheme 5 Synthetic applications of (*S*)-**1a**

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Supporting Information

Supporting information for this article is available online at https://doi.org/10.1055/s-0042-1751525.

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